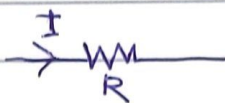


17.1) Given
 $I = 12 \text{ A}$
 $t = 1 \text{ min} = 60 \text{ s}$



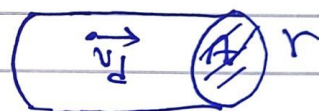
(a) $I = \frac{\Delta Q}{\Delta t} \Rightarrow \Delta Q = I \Delta t = 12 \times 60 = \boxed{720 \text{ C}}$

(b) $|q_e| = 1.6 \times 10^{-19} \text{ C}$

of electrons = $\frac{\Delta Q}{|q_e|} = \frac{720}{1.6 \times 10^{-19}} = 450 \times 10^{19}$
 $= \boxed{4.5 \times 10^{21}} \text{ electrons}$

17.5) Given $n = 5.8 \times 10^{28} \text{ e/m}^3$, $I = 10 \text{ A}$, $r = 10^{-3} \text{ m}$

$J = \frac{I}{A} = n q_e v_d$



$\Rightarrow I = n q_e v_d A \Rightarrow v_d = \frac{I}{n q_e A}$; $A = \pi r^2$

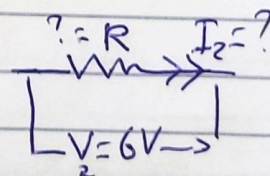
$v_d = \frac{10}{5.8 \times 10^{28} \times 1.6 \times 10^{-19} \times \pi \times (10^{-3})^2}$

17.10) Given $I = 10 \text{ A}$, $V = 2 \text{ V}$, $V_1 = 2 \text{ V}$, $I_1 = 10 \text{ A}$ Case 1

$V_1 = R I_1$

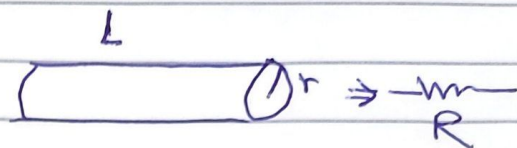
$V_2 = R I_2$

$\Rightarrow \frac{V_2}{V_1} = \frac{R I_2}{R I_1} = \frac{I_2}{I_1}$



$\Rightarrow V_2 I_2 = I_1 \frac{V_2}{V_1} = 10 \times \frac{6}{2} = \boxed{30 \text{ A}}$

17.11) Given, $r = 4 \times 10^{-5} \text{ m}$
 $L = 10 \text{ m}$



$$R = \rho \frac{L}{A} = \rho \frac{L}{\pi r^2}, \quad \rho \text{ from table 17.1} \\
= 1.47 \times 10^{-8} \Omega \text{ m} \\
R = 1.47 \times 10^{-8} \times \frac{10}{\pi (10^{-5})^2} = 0.468 \times 10^3 = 468 \Omega$$

17.12) Given $D = 10^{-5} \text{ m}$
 $\rho = 2 \Omega \text{ m}$
 $L = 0.3 \text{ m}$



$$A = \pi r^2, \quad D = 2r \Rightarrow r = \frac{D}{2} \\
= 0.5 \times 10^{-5} = 5 \times 10^{-6} \text{ m} \\
R = \rho \frac{L}{A} = 2 \times \frac{0.3}{\pi (5 \times 10^{-6})^2} = \frac{6}{\pi \times 25} \times \frac{10^{-1}}{10^{-12}} \\
= 0.076 \times 10^{11} = 7.6 \times 10^9 \Omega$$

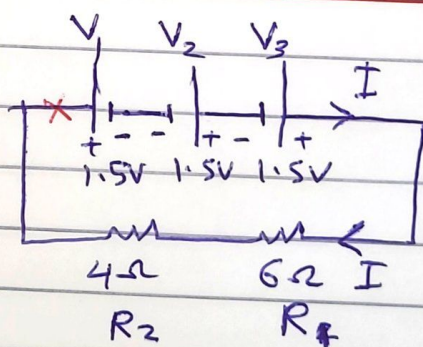
17.25) (a) $\sum V_{\text{loop}} = 0$

$$-V_1 + V_2 + V_2 - R_1 I - R_2 I = 0$$

$$-1.5 + 1.5 + 1.5 - 6I_1 - 4I_2 = 0$$

$$I_1 = I_2 = I$$

$$1.5 - 10I = 0 \Rightarrow I = \frac{1.5}{10} = 0.15 \text{ A}$$



(b) $V_{R_1} = R_1 I = 6 \times 0.15 = 0.9 \text{ V}$

$$V_{R_2} = R_2 I = 4 \times 0.15 = 0.6 \text{ V}$$

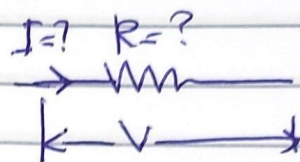
(P3)

17.34) Given $P = 1400 \text{ W}$, $V = 120 \text{ V}$

(a) $P = IV$, $V = IR \Rightarrow I = \frac{V}{R}$

$$\Rightarrow P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{(120)^2}{1400} = \frac{14400}{1400} \approx 10.29 \Omega$$

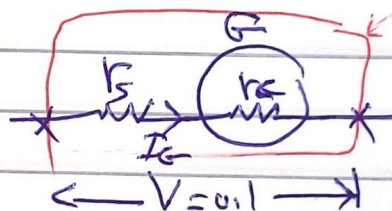
(b) $V = IR \Rightarrow I = \frac{V}{R} = \frac{120}{10} = 12 \text{ A}$



17.47) Given $R_G = 10 \Omega$, $(I_G)_{\text{max}} = 10^{-3} \text{ A}$, $V = 0.1 \text{ V}$

r_s is series

$$V = (I_G)_{\text{max}} r_s + (I_G)_{\text{max}} R_G$$



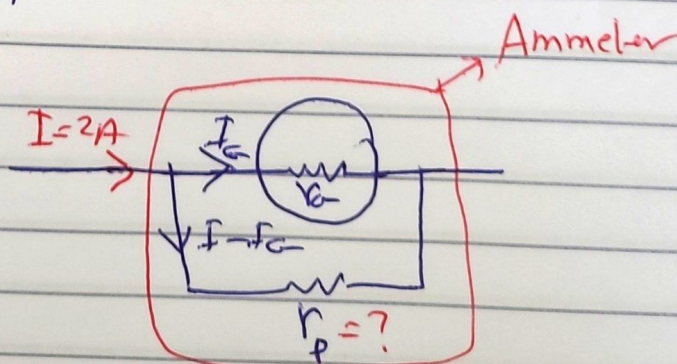
$$r_s = \frac{V}{(I_G)_{\text{max}}} - R_G = \frac{0.1}{10^{-3}} - 10 = 100 - 10 = 90 \Omega$$

NOTE: To convert the G to Ammeter measures 2A
We connect a resistance in parallel

$$\Rightarrow r_G (I_G)_{\text{max}} = r_p (I - (I_G)_{\text{max}})$$

$$\Rightarrow r_p = \frac{r_G (I_G)_{\text{max}}}{I - (I_G)_{\text{max}}}$$

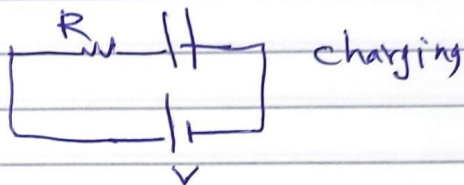
$$= \frac{10 \times 10^{-3}}{2 - 10^{-3}} = \frac{10^{-2}}{2 - 10^{-3}} \approx 0.5 \times 10^{-2} = 5 \times 10^{-3} \Omega$$



(P4)

17.50) Given $C = 10^{-5} \text{ F}$, $V = 100 \text{ V}$, $R = 10^3 \Omega$

(a) $\tau = RC = 10^3 \times 10^{-5} = 0.01 \text{ s}$



(b) $q_f = CV = 10^{-5} \times 10^2 = 10^{-3} \text{ C}$

17.51) Given $C = 10^{-4} \text{ F}$, $R = 100 \Omega$

(a) $q = q_i e^{-t/RC}$

$$e^{-t/RC} = e^{-2}$$

$$\Rightarrow \frac{t}{RC} = 2 \Rightarrow t = 2RC = 2 \times 100 \times 10^{-4} = 2 \times 10^{-2} \text{ s}$$

